TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	78.3 ± 2.45
Correlation Stability	83.3 ± 2.15
Distribution Consistency	88.2 ± 1.40
Entropy Preservation	91.1 ± 1.87
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	91.1 ± 1.64
Privacy Risk	53.4 ± 1.85
Data Density	100.0 ± 0.02
Coverage Score	10.7 ± 2.61
Feature Accuracy	78.8 ± 3.09
Novelty Score	0.4 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	claude-haiku-5-5
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	45.683	+0.936	2.1%	OK
TB	3.299	1.481	-1.818	55.1%	Drift
DB	1.486	0.595	-0.891	60.0%	Drift
Alkphos	290.576	235.492	-55.084	19.0%	Warn
Sgpt	80.714	45.177	-35.536	44.0%	Drift
Sgot	109.911	52.717	-57.194	52.0%	Drift
TP	6.483	6.619	+0.136	2.1%	OK
ALB	3.142	3.447	+0.305	9.7%	OK
A/G Ratio	0.947	1.073	+0.126	13.3%	Warn
Selector	1.286	1.460	+0.174	13.5%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	78.3 ± 2.45
Correlation Stability (*2)	83.3 ± 2.15
Distribution Consistency (*3)	88.2 ± 1.40
Entropy Preservation (*4)	91.1 ± 1.87
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	91.1 ± 1.64
Privacy Risk (*7)	53.4 ± 1.85
Data Density (*8)	100.0 ± 0.02
Coverage Score (*9)	10.7 ± 2.61

Feature Accuracy (*10)	78.8 ± 3.09
Novelty Score (*11)	0.4 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.659829059829059	0.599145299145299	0.654700854700854	-0.00512820512820513	
	9		7	25	
F1	0.761567486488495	0.686039770923495	0.761777094701435	0.00020960821293971	
	8	1	5	587	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.702564102564102	0.615384615384615	0.05641025641025642	*
		5	3		
F1	0.558743211938091	0.788360945310101	0.646244379040159	0.08750116710206815	*
	7	5	7		

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.705982905982905	0.678632478632478	0.688034188034188	-0.01794871794871795	
	9	7			
F1	0.812800269743901	0.756419397564493	0.778263466585712	-0.03453680315818943	*
	5	2			

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.702564102564102	0.702564102564102	0.704273504273504	0.00170940170940171	
	5	5	3	44	
F1	0.799790357154994	0.781762608390556	0.797393457239731	-0.00239689991526305	
	8	3	7	1	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.706837606837606	0.698290598290598	0.699145299145299	-0.00769230769230769	
	8	3	1	8	
F1	0.827909162452953	0.770394033752557	0.783625984587805	-0.04428317786514822	*
	7	4	4	4	

Model: AB

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6991452991452991	0.6888888888888889	0.014529914529914523	
F1	0.7821343746683586	0.779328569465419	0.77942351215357	-0.002710862514788581	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7051282051282052	0.7059829059829059	0.7102564102564103	0.005128205128205125	
F1	0.8091378556432121	0.7830987189226397	0.8063826992673416	-0.002755156375870577	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.6128205128205126	0.6957264957264957	-0.007692307692307676	
F1	0.8043780464360741	0.6512692747890649	0.7872072178320889	-0.017170828603985288	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6905982905982906	0.6863247863247863	0	
F1	0.7971299319768493	0.7654426210260246	0.7779379403604493	-0.019191991616400117	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6811965811965812	0.6760683760683761	0.7	0.018803418803418792	
F1	0.7877023987335553	0.7588388986082613	0.7942280581976299	0.006525659464074529	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	66	Male	15.2	7.7	356	321	562	6.5	2.2	0.4
2	4	Male	0.8	0.2	460	152	231	6.5	3.2	0.9
3	45	Male	0.8	0.2	140	24	20	6.3	3.2	1.0
4	73	Male	1.8	0.9	220	20	43	6.5	3.0	0.8
5	46	Female	4.7	2.2	310	62	90	6.4	2.5	0.6
6	51	Male	2.2	1.0	610	17	28	7.3	2.6	0.55
7	52	Male	0.9	0.2	212	28	31	6.8	3.4	1.0
8	34	Female	0.7	0.2	168	19	22	7.1	4.0	1.3
9	61	Male	2.1	0.9	298	54	67	6.3	3.1	0.9
10	45	Male	1.2	0.4	176	38	44	7.0	3.6	1.1
11	29	Female	0.8	0.2	145	22	18	7.4	4.2	1.3
12	70	Male	3.4	1.6	412	88	102	5.9	2.7	0.8
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	55	Female	0.8	0.2	155	21	17	6.9	3.8	1.4
2	64	Male	1.4	0.5	298	31	83	7.2	2.6	0.5
3	60	Male	0.5	0.1	500	20	34	5.9	1.6	0.37
4	38	Male	0.8	0.2	208	25	50	7.1	3.7	1.0
5	55	Female	10.9	5.1	1350	48	57	6.4	2.3	0.5
6	51	Male	2.9	1.2	189	80	125	6.2	3.1	1.0
7	52	Male	1.1	0.4	178	34	41	6.9	3.5	1.0
8	34	Female	0.9	0.2	162	27	25	7.6	4.1	1.2
9	61	Male	2.4	1.0	245	56	72	6.3	2.9	0.8
10	29	Male	0.7	0.2	140	31	28	7.8	4.4	1.3
11	47	Female	1.3	0.5	205	48	52	6.1	3.0	0.9
12	70	Male	3.1	1.5	290	88	110	5.8	2.6	0.8
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Male	0.8	0.2	367	42	18	5.2	2.0	0.6
2	42	Male	30.5	14.2	285	65	130	5.2	2.1	0.6
3	31	Male	0.6	0.1	175	48	34	6.0	3.7	1.6
4	30	Female	0.8	0.2	198	30	58	5.2	2.8	1.1
5	53	Female	0.8	0.2	193	96	57	6.7	3.6	1.16
6	60	Male	2.4	1.0	1124	30	54	5.2	1.9	0.5
7	45	Male	0.9	0.2	196	28	34	6.8	3.4	1.0
8	62	Female	1.3	0.4	234	36	41	7.0	3.5	1.0
9	29	Male	0.7	0.2	152	19	22	7.2	4.1	1.3
10	53	Male	4.6	2.1	402	88	131	6.1	2.6	0.7
11	38	Female	0.8	0.2	169	24	29	6.9	3.6	1.1
12	70	Male	1.5	0.6	248	41	57	6.4	3.0	0.9
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Female	0.9	0.3	189	23	33	6.6	3.9	nan
2	48	Female	1.4	0.8	621	110	176	7.2	3.9	1.1
3	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3
4	62	Male	0.6	0.1	160	42	110	4.9	2.6	1.1
5	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
6	46	Female	14.2	7.8	374	38	77	4.3	2.0	0.8
7	52	Male	0.9	0.2	187	24	31	6.8	3.5	1.0
8	34	Female	1.2	0.4	224	31	42	7.0	3.8	1.2
9	61	Male	2.4	1.1	412	58	96	5.9	2.8	0.9
10	47	Female	0.7	0.2	165	19	24	6.4	3.6	1.3
11	69	Male	3.8	1.9	540	72	118	6.1	2.5	0.7
12	29	Male	0.8	0.2	143	27	30	7.2	4.1	1.3
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	7.3	3.6	1580	88	64	5.6	2.3	0.6
2	21	Male	18.5	9.5	380	390	500	8.2	4.1	1.0
3	17	Male	0.9	0.2	279	40	46	7.3	4.0	1.2
4	40	Male	1.2	0.6	204	23	27	7.6	4.0	1.1
5	68	Female	0.6	0.1	1620	95	127	4.6	2.1	0.8
6	36	Male	2.8	1.5	305	28	76	5.9	2.5	0.7
7	45	Male	1.1	0.4	210	34	41	6.8	3.4	1.0
8	29	Female	0.7	0.2	165	18	22	7.2	4.1	1.3
9	61	Male	3.4	1.7	390	72	88	5.9	2.6	0.8
10	38	Female	0.9	0.3	178	26	24	6.9	3.6	1.1
11	52	Male	5.8	2.9	455	98	120	6.1	2.8	0.8
12	33	Male	0.8	0.2	172	30	35	7.4	4.2	1.3
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Female	1.2	0.4	358	160	90	8.3	4.4	1.1
2	32	Female	0.7	0.1	240	12	15	7.0	3.0	0.7
3	47	Female	3.0	1.5	292	64	67	5.6	1.8	0.47
4	48	Male	1.6	1.0	588	74	113	7.3	2.4	0.4
5	26	Male	6.8	3.2	140	37	19	3.6	0.9	0.3
6	17	Male	0.9	0.2	224	36	45	6.9	4.2	1.55
7	45	Male	0.9	0.2	189	28	31	6.8	3.4	1.0
8	33	Female	1.1	0.3	172	22	27	7.0	3.6	1.1
9	58	Male	2.4	1.1	298	64	89	6.1	2.8	0.8
10	27	Female	0.6	0.1	156	14	18	7.2	4.1	1.3
11	71	Male	4.9	2.2	412	97	133	5.9	2.5	0.7
12	39	Female	0.7	0.2	188	19	22	6.5	3.3	1.0
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58	Male	0.4	0.1	100	59	126	4.3	2.5	1.4
2	30	Male	1.6	0.4	332	84	139	5.6	2.7	0.9
3	38	Male	0.7	0.2	110	22	18	6.4	2.5	0.64
4	75	Male	6.7	3.6	458	198	143	6.2	3.2	1.0
5	48	Male	5.0	2.6	555	284	190	6.5	3.3	1.0
6	42	Female	0.8	0.2	182	22	20	7.2	3.9	1.1
7	54	Male	0.9	0.2	187	24	31	6.4	3.1	0.9
8	41	Female	0.7	0.2	172	19	22	7.0	3.8	1.2
9	63	Male	2.1	0.9	264	48	57	6.1	2.8	0.8
10	29	Male	1.1	0.3	198	33	36	7.3	4.0	1.2
11	70	Female	0.9	0.3	215	26	34	5.9	2.9	0.9
12	47	Male	3.4	1.6	305	88	74	6.7	3.3	0.9
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
2	48	Male	0.7	0.2	326	29	17	8.7	5.5	1.7
3	28	Female	0.9	0.2	316	25	23	8.5	5.5	1.8
4	32	Female	0.7	0.1	240	12	15	7.0	3.0	0.7
5	35	Female	1.0	0.3	805	133	103	7.9	3.3	0.7
6	35	Male	0.8	0.2	198	36	32	7.0	4.0	1.3
7	52	Male	0.9	0.2	212	31	36	6.8	3.4	1.0
8	34	Female	0.7	0.2	178	22	25	7.2	3.9	1.2
9	61	Male	1.4	0.5	290	44	58	6.1	2.8	0.8
10	29	Male	0.8	0.2	165	38	42	7.5	4.1	1.2
11	47	Female	1.2	0.4	245	27	33	6.9	3.2	0.9
12	70	Male	3.1	1.4	320	68	92	5.7	2.5	0.7
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	43	Female	0.9	0.3	140	12	29	7.4	3.5	1.8
2	45	Male	2.2	1.6	320	37	48	6.8	3.4	1.0
3	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
4	58	Male	1.0	0.5	158	37	43	7.2	3.6	1.0
5	38	Female	0.8	0.2	185	25	21	7.0	3.0	0.7
6	47	Female	3.0	1.5	292	64	67	5.6	1.8	0.47
7	52	Male	1.1	0.4	198	28	34	6.9	3.4	0.9
8	41	Female	0.7	0.2	172	19	22	7.1	3.8	1.2
9	63	Male	2.8	1.3	264	56	71	6.4	3.1	0.9
10	29	Male	0.9	0.2	148	34	29	7.6	4.3	1.3
11	57	Female	1.4	0.6	231	41	48	6.2	3.0	0.9
12	36	Male	3.6	1.9	305	88	112	5.8	2.7	0.8
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Female	0.7	0.2	170	21	14	5.7	2.5	0.7
2	75	Female	0.8	0.2	205	27	24	4.4	2.0	0.8
3	47	Female	0.8	0.2	236	10	13	6.7	2.9	0.76
4	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
5	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
6	33	Male	2.0	1.0	258	194	152	5.4	3.0	1.25
7	52	Male	0.9	0.2	201	28	31	6.4	3.2	1.0
8	44	Female	0.7	0.1	165	19	21	7.1	4.1	1.4
9	63	Male	3.1	1.4	245	88	102	6.0	2.8	0.9
10	29	Female	0.8	0.2	176	22	18	6.9	3.8	1.2
11	57	Male	1.2	0.5	218	46	52	6.6	3.5	1.1
12	35	Male	0.6	0.1	152	24	29	7.3	4.2	1.3
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3